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White paper



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Sport Is Pressure: The Participation Layer of Live Sports

This paper represents an early research perspective on interactive sports participation platforms.

Abstract

Live sports are defined by anticipation. Every attack, defensive error, and momentum shift carries the possibility of the next decisive moment. Fans do not experience matches passively. They constantly interpret the game, predicting what will happen next, debating decisions, and reacting to unfolding events in real time.

Despite this behavior, most digital sports products treat fans primarily as spectators. Broadcast platforms optimize for attention and advertising. Betting platforms optimize for transactions. Social media fragments discussion across feeds and comment threads. Few systems are designed around the actual cognitive experience of watching sport: anticipation, prediction, and reaction.

This paper proposes a new category in sports technology: the participation layer. Rather than consuming live sports passively, fans engage directly with the unfolding match by predicting moments and competing with friends and communities in real time. These interactions transform the instinctive interpretation fans already perform into structured participation.

Recent advances in live match data infrastructure, mobile interaction patterns, and AI systems capable of interpreting game context now make this possible at scale. Software can detect moments of uncertainty during a match and generate contextual prediction prompts, turning natural fan instincts into interactive competitive experiences.

The result is a new category of sports product built not around viewing the match, but around participating in it.

The next generation of sports platforms will not simply show the game.
They will allow fans to play along with it.

The Thesis

For decades, the sports technology industry has optimized for two things: distributing content to more people, and extracting transactions from the fans who show up. Broadcast platforms competed on reach and production quality. Betting markets competed on liquidity and convenience. Fantasy platforms extended engagement across the season. Each of these systems was built around the same assumption: that fans are fundamentally an audience, and the role of technology is to serve that audience more efficiently.

This framing overlooks the defining cognitive behavior of sports fans. Watching a live match is not a passive act. Fans constantly anticipate what will happen next. They read momentum shifts, track individual matchups, notice subtle tactical adjustments, and form expectations about how the next moment will unfold. This prediction instinct operates continuously beneath the emotional experience of sport. It happens before the goal, before the foul, before the substitution. It is the hidden cognitive activity running underneath the match.

In practice, fans are already playing a prediction game in their heads. They argue with friends about what will happen next. They react to missed expectations. They reinterpret earlier moments as the match evolves. This dynamic creates a constant loop of anticipation and resolution that defines the rhythm of live sport.

The participation layer is a system that structures this instinct. It takes the natural prediction behavior of fans watching a live match, makes it explicit and competitive in real time, and records the outcome of those predictions over time. Instead of treating fans only as spectators, the system treats them as interpreters of the game. The result is a new kind of sports product: one built not around watching the match, but around participating in it.

This layer also creates something that does not exist in sports technology today: a structured record of how fans interpret the game as it unfolds. Match by match, moment by moment, the system captures the predictions fans make and the accuracy of those predictions over time. What emerges is a behavioral layer that reflects how people read sport in real time.

Building this layer requires three technical conditions to exist simultaneously: live structured match data at low latency, AI systems capable of interpreting match context, and mobile platforms capable of delivering interaction during the match itself. For most of the history of sports media, these conditions did not exist together.

Today they do.

The participation layer is therefore not a distant concept. It is a technical layer that has only recently become possible, and one that has not yet been built at scale.

01

Introduction

Why the fan experience is richer than any product built for it

Sport is a pressure system. It generates anticipation continuously, compresses it into discrete moments of resolution, and then starts again almost immediately. This is what separates it from scripted entertainment. You cannot skip to the end. Every moment carries genuine stakes, and the fan watching is not simply an observer. They are constantly predicting what comes next.

There is a moment in every close match, somewhere in the final twenty minutes, when the atmosphere changes. Not just the noise. The quality of attention in the stadium shifts. Players feel it. Managers feel it without wanting to show it. The game has tilted, and everyone present knows it before any commentator names it.

Fans sense this before anyone articulates it. Not from statistics, but from years of watching. The intuition is fast, involuntary, and emotionally charged. It is one of the defining experiences of live sport. Almost no technology product has been built around it.

Technology revolutions rarely replace existing systems. They introduce a new layer that changes how people interact with everything beneath it. The internet created the application layer. Mobile created the personal computing layer. Each new layer did not destroy what came before. It opened a different relationship with it. Live sports are now ready for their own new layer. Not a replacement for broadcast. Not a competitor to betting or fantasy. A participation layer, built directly on top of the live match experience that already exists.

The next generation of sports products will not be built around content or betting.
They will be built around participation during the match.

The sports technology industry has produced impressive things. Live data feeds with sub-second latency. Fantasy platforms with enormous user bases. Betting markets processing hundreds of billions in volume every year. These products share a fundamental misreading of the fan experience. They are built for spectators. They optimize for consumption.

None of them capture the predictive experience of watching a live match. None let a fan commit to what they think happens next, right now, in this match, and find out within the minute whether they were right.

Fans already do this instinctively. In living rooms, group chats, pubs, and stadium stands, they predict, argue, and react constantly. The technology simply has not caught up to the behavior.

This paper examines that gap: why it exists, why it matters, and what the participation layer of live sports looks like as a system.

02

Sport Is Pressure

The psychological architecture of live competition, and the prediction instinct it generates

Strip away the teams, the histories, and the tribalism, and live sport is fundamentally a pressure machine. It generates tension continuously, releases it in discrete moments, and begins again. The outcome is always unknown. That unknowing is the entire point.

The neurological experience of watching live sport is well-documented. Cortisol and adrenaline rise during high-intensity periods. Dopamine releases at the moment of resolution: the goal scored, the penalty saved. The physical response is not metaphorical. Heart rate data from supporters during matches consistently mirrors the profile of moderate physical exertion.

What is less often examined is the cognitive experience running underneath the emotional one. Fans are not simply feeling. They are constantly thinking. Anticipating. Simulating probable futures. This activity is involuntary, continuous, and it is the actual substance of fan engagement with a match.

Momentum as a cognitive concept

Momentum is used loosely in commentary, but fans track it with real precision. A team that has dominated the last ten minutes without scoring has created pressure that most fans sense before any analyst names it. Possession statistics and shot counts formalise something fans already perceive instinctively: the balance has shifted.

This pattern recognition is active, not passive. When a striker runs in behind a flat defensive line, a fan does not simply observe the movement. They form an expectation about what comes next. The brain is running a simulation. That simulation is the hidden cognitive activity inside every live match, and it is the thing that the participation layer is built to surface.

The moment before the moment

Anyone who has played competitive sport knows a specific feeling: you are already moving before you have consciously decided to. The best players are not faster because they react more quickly. They are faster because they predicted the situation before it arrived. They read the game.

Fans experience a cognitive version of this from the stands. The half-second before a corner is delivered when you are already leaning toward the near post. The held breath before a penalty. These are not passive responses to events. They are anticipatory predictions, arriving slightly before the events they concern. Sport creates this experience continuously. Almost no product has been built around it.

The hidden prediction game

There is a phrase in poker for the game happening beyond the cards: the meta-game. Live sport has its own version. While the physical match unfolds on the pitch, another game plays simultaneously in the minds of everyone watching. A centre-back already on a yellow card with an hour to play. The opposition striker pulling him wide, testing his patience. Every fan is running a quiet probability model: how long before he cracks? What does the manager do? These are sustained cognitive tasks, and they are part of why people cannot look away.

The cognitive load of watching a live match is substantially higher than watching a recording of the same match. Remove genuine uncertainty and the mental engagement collapses.

This was formalised in research by Bluma Zeigarnik, whose work on memory demonstrated that unresolved tasks retain a stronger cognitive hold than completed ones. In sports the mapping is almost exact. Open questions, whether he scores, whether the referee intervenes, whether they hold the lead, accumulate and sustain attention until the match resolves them. It is a fundamental property of human cognition, and live sport exploits it continuously.

The social layer of prediction

Watch a group of people viewing a tense match and you will see this hidden game break into the open. Someone says "he is going to get sent off" before the second yellow comes. Someone else calls the substitution. When the corner is won, two people simultaneously name the same player before the delivery. This informal prediction game is universal. It happens across football cultures and languages, in homes and stadiums, between strangers in pub seats. It is not prompted by any product. It emerges spontaneously from the predictive experience that fans are already having individually.

The significance is straightforward. The behavior that a participation layer depends on is not hypothetical. It already exists, happening informally alongside every live match watched by more than one person. The product category does not need to create the behavior. It needs to give it somewhere structured to go.

Game sense and the prediction instinct

In competitive gaming, there is a concept called game sense: the ability to anticipate what will happen next by reading the state of play and projecting its likely development. High-level players in games like Counter-Strike or League of Legends are distinguished less by mechanical speed than by the quality of this instinct. Sports fans demonstrate the same thing. The supporter who watches a defensive shape and knows before anything happens that the team is vulnerable to a specific run. The cricket watcher who tracks a bowler's pattern and predicts the next delivery. This anticipatory reading is one of the most valued forms of knowledge in sports culture, and there is currently no system that captures it.

Football intelligence is one of the most culturally valued forms of knowledge in most societies on earth. Almost none of it is formally captured anywhere.

03

The Prediction Behavior of Fans

How the instinct already exists, and why it has no structured outlet

Between sixty and eighty percent of fans under 35 use a second screen while watching live sports. This figure has been stable long enough to no longer be a trend. It is the baseline condition of watching sport for a large part of the audience.

The more interesting question is what the second screen is actually being used for. Group messaging, social media reactions, statistics lookups. None of these were designed for live sports co-viewing. They are workarounds: fans building an interactive experience from tools never intended for it.

The group chat as prototype

The football group chat is possibly the most under-examined product in sports technology. In a close match it runs at the pace of a trading floor. Predictions appear before moments occur. Reactions are instantaneous. Arguments about decisions open, briefly resolve, and reopen. Informal scores of who called what correctly emerge on their own. The chat is, in effect, a rough prototype of the participation layer: built spontaneously, from messaging infrastructure never intended for it.

What the group chat lacks is structure. Predictions are not timestamped against match events. There is no persistent record. At the end of a season, no one knows who the group's most accurate predictor was across thirty-eight matchweeks. The informal game is happening. Its outputs dissolve into the conversation.

Second-screen usage during sport does not distribute evenly across a match. It spikes around events: goals, red cards, VAR decisions. It also rises in the build-up to high-tension phases. Fans reach for their phones before moments as well as after them. They are not only reacting. They are anticipating, and they want somewhere to put that anticipation. What they are reaching for, without an established name for it, is the participation layer.

The fan participation model

Any product built around fan prediction behavior has to be grounded in an accurate model of how fans actually experience live sport. The following describes five phases that any honest fan will recognise.

The Core Equation

Sport = pressure

Pressure = anticipation

Anticipation = prediction

Prediction = the participation layer

The Fan Participation Loop

Anticipation
↓
Prediction
↓
Commitment
↓
Reaction
↓
Reputation

Each phase already happens informally, in group chats and stadium conversations, during every match. The participation layer gives it structure, persistence, and social consequence.

Anticipation: the match develops, a situation builds, and fans form expectations without being prompted. Something is coming. They are constructing probable futures from what they are seeing.

Prediction: from anticipation, the fan forms a specific position. Not just a feeling that something will happen, but a view on what will happen. Will the striker score? Will the booking come? The prediction is still internal, but it is already specific.

Commitment: the prediction is stated, out loud or on screen. This changes the psychological relationship with the outcome. People who state a prediction experience **stronger emotional**

reactions to its resolution than people who held the same prediction privately. The fan who typed "he's getting sent off" is more engaged in the next five minutes than the fan who simply thought it.

Reaction: the match resolves the uncertainty. For the fan who committed, this moment carries a second layer: was I right? The reaction is instantaneous. Vindication and disruption both produce stronger engagement than passive observation.

Reputation: if there is a record, and others can see it, something accumulates. Casual prediction behavior happens constantly but leaves no trace. A fan's accuracy across a season of matches, their record of reading the game correctly, has nowhere to live. That absence is the market gap.

From Instinct to System

What this loop reveals is that participation in sport already exists. It happens informally in stadium seats, living rooms, group chats, and comment threads. Fans constantly anticipate, predict, react, and defend their interpretations of the match. But today these interactions leave no structure behind. The insights fans generate while watching the game disappear the moment the match ends.

There is no persistent system that captures how fans read the game over time. The participation layer changes this dynamic. By turning anticipation into explicit prediction and recording outcomes across matches, it creates a structured record of how fans interpret live sport.

Over time, this record becomes a new kind of sports data: a map of collective fan intelligence.

Instead of measuring only what players do on the field, the system also captures how millions of fans understand the game as it unfolds. This is the foundation of a new category of sports technology, one built not around watching sport, but around participating in it.

04

The Participation Layer

A new category in sports technology, and why existing products have not served it

The term participation layer borrows from software architecture, where it describes the component of a system that sits between infrastructure and the end user: the interface through which human intention meets machine capability. In sports, the analogy is direct. Broadcasting is infrastructure. It delivers the match to the fan.

The fan has intentions, predictions, reactions, judgments, that have no structured outlet within the experience. A participation layer sits between the broadcast and the fan's active psychological engagement, converting informal, individual, and ephemeral prediction behavior into explicit, social, and persistent competition.

For decades, sports technology focused on the top of this stack: how to distribute content to more people, more efficiently, with richer production value. The participation layer asks a different question. Not how to deliver the match, but what the fan does with what they are experiencing. Distribution is solved. Engagement architecture is not.

Where sports technology sits today:

Product	Optimizes for	Relationship with the match
Broadcast	Attention	Passive viewing
Betting	Transactions	Financial stake, pre-match
Fantasy	Season stats	Disconnected from live moments
Participation Layer	**Participation**	**Real-time prediction during live moments**

The participation layer is not a variation of any existing category. It is what is missing from the table above.

What it is not

It is not a statistics app. Statistics describe what has happened. A participation layer is about what will happen next. These are categorically different cognitive experiences.

It is not a fantasy product. Fantasy requires pre-match setup and resolves on a cadence entirely disconnected from live match events. A participation layer operates in the same time window as the sport itself: minutes, sometimes seconds.

It is not a betting product. There is no financial stake. The currency is knowledge, judgment, and reputation within a community that cares about the same outcomes you do.

What betting got right, and missed

Betting platforms understand, better than anyone else in sports technology, that fans want their predictions to matter. The global sports betting market exceeds two hundred billion dollars annually. That is not a market driven by financial return. It is a market driven by people who want to be right about what happens next in sport.

The betting industry has mapped the territory. It has not served it well. The financial layer is not the point. The prediction is the point.

What makes the participation layer possible now

Fans predicted outcomes informally before smartphones existed. What has changed is the technical capability to build it at scale. Three conditions have converged. Live structured match data is available with sub-second latency, covering thousands of matches across dozens of competitions. AI systems can now interpret that data contextually in real time. And mobile device penetration among the primary sports fan demographic means the delivery mechanism is already in place. The phone is in the fan's hand. It simply needs something worth doing with it.

Figure 2: The Sports Ecosystem Model



The participation layer sits between the existing broadcast infrastructure and the fan's active predictive engagement. It is not a replacement for any existing layer. It is the layer that has been missing.

05

The Participation Stack

How the system transforms passive viewing into active participation

The participation layer is not a feature. It is a stack of technical and behavioral systems working together. Understanding how each layer functions helps explain both why this has not been built before and why it is now possible.

Think of the AI layer as a real-time interpreter of the match. It does not simply detect events. It translates raw match events into moments of prediction that fans can engage with, taking the structured data flowing in from the pitch and surfacing the narrative significance embedded in it.

Reading a match event correctly requires understanding its narrative significance.

Not just detecting that a yellow card has been shown, but knowing that this card, shown to this player, in this competition, at this scoreline, is a crisis for one side and an opportunity for the other.

Figure 1: The Participation Stack



Figure 1. The participation layer structures fan predictions around real-time match events. Each layer depends on the one above it. The fan decision is only meaningful if the prediction window is contextually relevant. The prediction window is only relevant if the AI interpretation is accurate. The AI interpretation is only accurate if the match context signals are properly modeled.

Live match events and context signals

Live sports produce a continuous stream of discrete events: shots, fouls, substitutions, cards, goals. Each event is structurally identical in data terms, a timestamp, an event type, a player, a team. But their narrative significance varies enormously depending on context.

A goal in the 90th minute that makes it 4-0 in a dead rubber is, narratively, almost meaningless. The same event in a title decider is among the most significant things that can happen in a season. A system that cannot distinguish between these two situations will generate prompts that feel irrelevant to anyone watching.

This is why the match context signals layer exists. It holds not just events, but the state of the match: the score, the competition stakes, the time remaining, the tactical situation, and the accumulated tension of what has happened before. The participation stack cannot function without an accurate model of match state.

The AI interpretation layer

Recent AI systems, specifically large language models trained on substantial volumes of sports commentary, tactical analysis, and match reporting, can interpret match context with real precision. Through training they have absorbed the contextual logic of what makes a match event significant. They understand, in a functional sense, that a late match situation in a title race generates fundamentally different tension from the same event in a pre-season friendly.

The AI interpretation layer evaluates the current match state and determines whether a prediction window should open. The system is not selecting from pre-written questions. It is generating a question specific to this match moment: these players, this score, this competition, this trajectory.

Prediction windows and timing

A participation layer only works if prompts arrive when the fan's attention is already engaged with the prediction it describes. A question about whether a player will be shown a red card is only meaningful when the player is already on a booking in a tense match, not five minutes after the incident has resolved. The time between a qualifying match event occurring and a contextual prompt arriving on a fan's device must be measured in seconds.

The defensibility of the system

As fans participate during matches, the platform accumulates a dataset that no other system generates: fan prediction behavior mapped against real match contexts. Every answer is a data point. Which question was asked, in which match situation, and how the fan responded.

Over time, this creates a behavioral layer linking match context, prediction patterns, and reputation systems. The AI interpretation layer becomes more precisely calibrated as it learns which types of match situations generate genuine engagement and which merely appear significant. Better calibration produces better prediction windows.

Better windows drive more participation. More participation generates more data. Combined with AI models trained specifically to interpret match context and generate prediction windows, this creates a feedback loop between match data and fan behavior.

The calibration data cannot be acquired without equivalent live user traffic. You cannot buy it. You have to earn it, match by match.

Industry infrastructure already in place

The market infrastructure for a participation layer already exists in large part. Genius Sports, which holds official data distribution rights for the NFL, the Premier League, and dozens of other major competitions, went public in 2021 at a valuation exceeding one billion dollars. Stats Perform, Sportradar, and a constellation of regional data providers have collectively attracted billions in institutional capital over the past decade. The premise underlying all of it is the same: live sports data, delivered with low latency and structured for machine consumption, is extraordinarily valuable.

These platforms do not primarily generate revenue by delivering data to fans. They generate revenue by delivering data to sportsbooks. The real-time event stream from a live Premier League match is the raw input for the global in-play betting market. The infrastructure is fast, reliable, and comprehensive because billions of dollars of betting decisions depend on it every week.

The infrastructure for a participation layer has already been built. It is powering the wrong product.

The real-time data infrastructure that a participation layer needs has already been built and paid for, primarily by the betting industry. The broadcast infrastructure to reach fans at scale has been in place for decades. The mobile devices through which a participation product would be delivered are already in fans' hands during matches. What does not exist is the layer between these systems and the fan's predictive engagement.

Platform implications

A mature participation layer does not just add interactivity to live sport. It creates the conditions for a new kind of sports platform, one built around identity, community, and competition rather than content delivery.

A fan who has participated across an entire season of matches has built something real: a record. Not a follower count, not a transaction history, but a documented log of how they engaged with the game, which moments they engaged with, and whether their predictions proved accurate. Over time this becomes a form of sports identity that no other platform currently generates.

Squads, leagues, and leaderboards built around prediction accuracy create a layer of community competition that sits naturally on top of the match itself. The social experience of watching sport together is already one of its most important properties. The participation layer extends this across time and across matches. As the participation layer matures, it creates natural entry points for fan creators and analysts. A pundit with a strong prediction record has a demonstrable claim to credibility. A fan account that correctly called the tactical shift three matches in a row has a form of social proof that no engagement metric currently captures. Value in participation platforms accumulates in three places: the social layer, the relationships and rivalries that form over time; the identity layer, the persistent record of how accurately a fan reads the game; and the data layer, the behavioral signal generated by millions of fans committing to predictions in real time. None of these are advertising-dependent. None require a transaction. They are created by participation itself.

06

Conclusion

The prediction was always there. Technology gives it a place to live.

Sport is a pressure system. It generates anticipation continuously, resolves it in discrete moments, and starts again almost immediately. Fans experience this not as spectators but as predictors. They form expectations, commit to positions, and have those positions validated or contradicted by events that are genuinely uncertain.

This behavior is universal. It happens in every living room, every group chat, every stadium in every sport on earth. Fans already predict outcomes constantly during matches. They debate, react, and compare instincts with the people watching alongside them.

They do this without being asked, without a product to structure it, without any way to record or share the result. What has been missing is a system that captures this behavior and structures it in real time. Not betting. Not fantasy. Not statistics. A participation layer: one that connects watching the match with participating in it, that converts the natural prediction instinct of a fan into something competitive, social, and persistent.

The sports data industry has already built the infrastructure. The betting industry has proven the demand. Esports has validated the format with the right demographic. Mobile devices are already in every fan's hand during a match. The AI systems capable of interpreting live match events contextually, in real time, now exist. Each condition developed independently. Together they create the environment in which the participation layer becomes possible.

The next generation of sports products will not simply show the game.

They will allow fans to participate in it.

Pulse is building the system that captures this layer. A structured record of how fans interpret the game, generated match by match.

https://joinpulse.live/

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